Inventions and innovations

Answers

1. Use a dictionary to look up the words ‘invention’ and ‘innovation’. These words don’t mean quite the same thing, do they? Describe the difference between these terms in your own words.

The term ‘inventing’ means to create or invent something new, whereas ‘innovation’ means to introduce something that is new. An invention can be a new innovation.

2. What is the difference between ‘science’ and ‘technology’?

Science is a way of experimenting, observing and explaining natural phenomena, whereas technology is the application of science.

3. The picture at right shows what is believed to be the earliest known earthquake detector or ‘seismometer’. It was invented by the Chinese scientist Zhang Heng almost 2000 years ago, and was able to detect earthquakes 854 kilometres away.



Photo: Public domain

Suggest how the seismometer may have worked.

Answers may vary but could be similar to the following. Perhaps when an earthquake hits, something may fall from the dragons’ mouths into the open mouths of the frogs. The more that fall, the greater the intensity of the earthquake. Direction could also be determined.

4. List what you consider to be the five most important inventions in the past 100 years.

Answers will vary but could include the following.

Many of the inventions that we use today — such as the telephone, automobile, camera, light bulb and sewing machine — were invented more than 100 years ago.

Inventions in the last 100 years include penicillin, computers, satellites, spaceships, aeroplanes, microchips, the internet, CDs and DVDs.

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Clocks

In the past, church bells or a town crier helped communities to know the time. Clocks were invented and placed on public buildings in western Europe from the end of the thirteenth century. By the seventeenth century, smaller home versions were available. These clocks were not very accurate because there was no way of controlling the rate at which they worked.

It was Galileo (1564–1642) who first noticed that a pendulum swinging backward and forward kept equal time on each swing. Galileo made this discovery during a church sermon as he watched a swaying chandelier and used his pulse to time its swings. The Dutch scientist Christiaan Huygens (1629–1695) applied Galileo's discovery to control the rate of clocks, inventing the pendulum clock in 1656.

The discovery of the pendulum as a method of recording accurate time solved the problem of how to measure longitude. It became possible to explore the world. Trade between nations began and Europeans started to colonise distant lands.

What would your everyday life be like if man-made clocks had never been invented? Write a paragraph below describing what would change in your typical day if that were true.



Photo: Vadim Yerofeyev / 123RF

Answers will vary.

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